Self-Supervised Isotropic Resolution Enhancement and Denoising Diffusion for 3D Abnormalities Detection from Anisotropic MRI

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Highlights

- Self-supervised GAN–diffusion pipeline yields isotropic 3D lesion maps from 8× anisotropic scans.
- Outperforms state-of-the-art with better perceptual metrics and favored by orthopedist review.
- Boost anatomy & lesions delineation; robust across multi-centers, applied to different protocols.

Abstract

Accelerating musculoskeletal MRI while preserving diagnostic detail remains challenging because acquiring fully-isotropic ground-truth volumes is clinically costy. We present a two-stage, fully self-supervised pipeline that learns directly from anisotropic scans—obviating any paired high-resolution data—and converts highly anisotropic (8×) turbo-spin-echo volumes into isotropic images and 3-D abnormality maps. Stage 1 employs a multiview GAN with contrastive and adversarial objectives to restore through-plane resolution; Stage 2 leverages an anatomy-conditioned denoising-diffusion model to inpaint healthy tissue, yielding voxel-wise lesion maps without external annotations.

On 2,225 Osteoarthritis Initiative knee scans from five different imaging centres, the framework reduced Fréchet inception distance from 407.4 → 254.4 (coronal) and 429.9 → 266.9 (axial), achieved the best KID/LPIPS scores among competing unsupervised methods, and was preferred in 65–67 % of blinded orthopaedist comparisons. Crucially, isotropic enhancement propagated to downstream tasks: femur–tibia segmentation F1 scores increased and previously confluent bone-marrow lesions were separated into discrete entities, enabling precise volumetric quantification. Robustness experiments demonstrated consistent gains across five imaging centers, synthetic noise/contrast perturbations and two external protocols (T2-SPACE spine, DESS knee), underscoring broader generalisation.

By eliminating the need for ground-truth isotropic images while surpassing state-of-the-art unsupervised super-resolution in both perceptual quality and clinical utility, our method offers an immediately deployable solution for retrospective cohort study and prospective scan-time reduction, paving the way for broader MRI applications in heterogeneous clinical environments.

Keywords: Self-supervised learning, Denoising diffusion model, Isotropic enhancement, 3D Abnormality detection, Musculoskeletal imaging,

1. Introduction

High-resolution magnetic resonance imaging (MRI) plays an essential role in modern clinical diagnosis and treatment planning procedures because it produces detailed images of anatomical structures and pathological changes. However, the acquisition of truly isotropic high-resolution three-dimensional (3D) MRI volumes is restricted by prolonged scan times, motion artifacts, and hardware limitations; therefore, clinicians often settle for more time-efficient anisotropic acquisition methods, which have substantially lower through-plane (between-slice) resolutions than in-plane (within-slice) resolutions (Fritz et al., 2016; Kijowski et al., 2017; Mugler, 2014; Subhas et al., 2011). Such anisotropic acquisition techniques adversely affect the resulting diagnostic accuracy and hinder automated image analysis algorithms, which typically assume a consistent spatial resolution across all dimensions (Kijowski et al., 2009; Zhai et al., 2005). Musculoskeletal imaging for osteoarthritis (OA), specifically that of the knees, necessitates the use of high-resolution 3D MR images with equal levels of detail in all directions because the knee consists of different tissues (e.g., articular cartilage, subchondral bone, and ligaments), each of which is best visualized from a different orientation (Peterfy et al., 2008). These different tissues often display distinct pathological changes, such as cartilage losses or bone flattening effects, which are best appreciated within a 3D space. Musculoskeletal MR images commonly exhibit substantial anisotropy, which is driven by clinically necessary tradeoffs between the scanning time and slice thickness. For example, large multicenter cohorts, such as the Osteoarthritis Initiative (OAI), contain multimodal MRI data with in-plane/through-plane resolution ratios of up to approximately 9.6 (Peterfy et al., 2008). It is therefore difficult to detect abnormalities in planes that are perpendicular to the primary acquisition direction, especially when subtle abnormalities are only visible across a few slices. Moreover, the large slice thickness values in anisotropic MR images make it challenging to study the progression of diseases because subtle pathological changes may be missed if they do not span multiple slices.

Recently, extensive research has been conducted on MR image resolution enhancement methods that generate isotropic, high-resolution volumes from low-resolution input data (Chaudhari et al., 2020; Saitta et al., 2024; Song et al., 2023; Zhu et al., 2017). Although deep learning models have shown considerable potential for achieving this goal, their clinical impact remains constrained by the scarcity of high-resolution and low-resolution training images (Qiu et al., 2023), especially in retrospective studies where historical MRI data often exhibit heterogeneous anisotropy and degradation patterns. This scarcity of paired high- and low-resolution datasets can force researchers to incorporate simulated processes or degradation processes learned from independently trained models into their own models, potentially limiting their generalizability (Zhao et al., 2021). To address these challenges, researchers have resorted to self-supervised superresolution frameworks that leverage external information, such as high-resolution images derived from different modalities (Iwamoto et al., 2023; Kang et al., 2024; McGinnis et al., 2023; Wang et al., 2023a, 2023b; Zhang et al., 2024), multiple acquisition methods (Gundogdu et al., 2024), k-space (Tsai et al., 2024) or simulation data (Iglesias et al., 2021).

To compensate for the lack of paired high-resolution data, recent research efforts have implemented self-supervised learning through multistage or multi-generator frameworks that rely on the cyclic consistency loss (Liu et al., 2021a, 2021b; Sun et al., 2023; Xie et al., 2022). For example, a two-stage framework that involves first constructing a synthetic high-/low-resolution paired dataset from anisotropic MR volumes and then using this dataset to train a network for arbitrary-scale superresolution processes was developed (Remedios et al., 2023). Zhang et al. designed a self-supervised superresolution framework in which downsampled high-resolution in-plane slices are used for model training and 3D volumes are reconstructed by averaging two-dimensional (2D) orthogonal predictions (Zhang et al., 2023). In another study, a cycle generative adversarial network (GAN)-based architecture was first used to generate low-resolution data from a high-resolution source for domain correction purposes, and then a second-stage network was trained using the assumed high-/low-resolution pair for image upsampling (Zhou et al., 2023). Similarly, a two-stage self-supervised cycle consistency network was first pretrained on synthetic low-/high-resolution pairs derived from sagittal and coronal slices, which were then refined by imposing cycle consistency constraints on axial triplets (Lu et al., 2021).

Despite these advances, the existing self-supervised methods are limited in scope. They rely either on simulated distributions of low-resolution data obtained by generating low-resolution data from high-resolution in-plane slices through synthetic degradation or on weaker self-reconstruction constraints, which frequently introduce image artifacts. In addition, their multistage training strategies introduce additional computational overhead and optimization challenges, and their use of 2D slice-based processing rather than true 3D architectures may lead to the suboptimal preservation of volumetric anatomical continuity. Finally, these methods are focused primarily on moderate anisotropic resolution ratios (typically ~4:1), limiting their applicability to clinical MRI acquisition scenarios, which frequently exhibit much higher degrees of anisotropy.

Despite the growing interest in MR image superresolution techniques, relatively few studies have explicitly investigated how improved isotropy might enhance the process of delineating abnormal imaging features in a fully 3D context. Recent advances in generative models, particularly denoising diffusion models, have shown potential for detecting abnormalities or abnormal patterns in MR images (Fontanella et al., 2024; Kascenas et al., 2023; Sarkar et al., 2025; Wyatt et al., 2022; Zimmerer et al., 2022); however, the application of 3D generative models for synthesizing volumetric MR images remains computationally intensive, which limits the potential utility of these methods for extensive longitudinal studies because of the large number of cases needed. Therefore, investigating how resolution-enhanced isotropic reconstructions can improve the computational efficiency of models in terms of detecting volumetric anomalies in OA knees is of particular interest. Improvements in this area would enable more accurate characterizations of joint abnormalities and diagnoses based on large longitudinal cohort studies.

In this study, we propose a new framework that combines a self-supervised resolution enhancement GAN and a denoising diffusion model to enhance anisotropic knee MR images and facilitate the 3D isotropic detection of knee abnormalities. The self-supervised GAN transforms the original anisotropic volumes into high-resolution isotropic volumes, whereas the diffusion model inpaints the abnormal regions contained in each 2D sagittal slice to generate synthetic "normal" tissue. By contrasting the healthy and abnormal 3D reconstructions, our approach enables abnormal regions to be more accurately localized and characterized. We demonstrate that our approach yields better image quality than the current superresolution methods do, as evidenced by quantitative assessments and expert reviews. In particular, the enhanced isotropic resolution improves the ability to visualize the abnormal features of the knee across multiple anatomical planes, thereby maximizing the diagnostic utility of the proposed method even in cases with high degrees of anisotropy. We demonstrate improved image quality, anatomical continuity, and clinical utility through extensive quantitative evaluations and expert radiological assessments, highlighting the potential utility of the model in OA research.

2. Method

2.1 Overall Model Framework

We propose a two-stage framework for enhancing anisotropic MRI scans to isotropic resolutions and subsequently detecting 3D abnormalities using a denoising diffusion model, as illustrated in **Figure 1**. The first stage employs a self-supervised multiview GAN to enhance the original anisotropic image (x_{2D}) to a high-resolution anisotropic image (x_{3D}), mitigating the challenges arising from the absence of paired high-resolution data. The second stage leverages a denoising diffusion model with anatomical control to delineate and isolate abnormal features, resulting in healthy counterfactual images (y_{2D}). Finally, 3D abnormalities (a_{3D}) are assessed by comparing the enhanced diseased volumes with their healthy counterparts (x_{3D} and y_{3D}, respectively), both of which are enhanced to isotropic resolutions by the GAN model.

2.2 Data Preprocessing

We used sagittal intermediate-weighted turbo spin–echo (IW-TSE) MR images that exhibited significant anisotropy between their in-plane resolution (0.357 mm × 0.357 mm) and through-plane resolution (3.0 mm slice thickness), which we collected from the baseline and follow-up cohorts studied by the OAI. During model training, the 3D MR images were divided into 16 × 128 × 128 voxels image subvolumes and then resampled to a final voxel size of 128^3, corresponding to a volume matching the anisotropic ratio of 8 of the original scan. We excluded images with artifacts and ensured that the subjects exhibited consistent anatomical alignment. The study cohort consisted of participants with significant contralateral differences in the OA symptoms between the knees, as assessed by the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) pain scale. We identified subjects who reported one knee to be painful (WOMAC ≥ 4, cases) and the other to be pain-free (WOMAC = 0, controls). Finally, 2225 paired knee MR images were used for model training and validation purposes, with no subject overlap between the sets. Further details regarding the subject selection criteria and data preprocessing strategies are provided in Supplementary Material A.

2.3 Isotropic MR Image Enhancement

The first component of a self-supervised GAN model (**Figure 1a**) transformed the anisotropic MRI volumes of the case and control images to isotropic resolutions, enabling a precise comparison between their diseased regions. To enhance the resolutions of the images, we converted the low-resolution coronal and axial views to high-resolution sagittal views through self-supervised learning, which eliminated the need for high-resolution ground-truth scans or direct estimation of the degradation function. We adopted a 3D encoder–decoder design featuring three downsampling blocks in the encoder, with each block containing two consecutive 3D layers followed by a 3D batch normalization layer and a downsampling operation. The number of feature channels doubled at each downsampling step, while the decoder symmetrically upsampled the spatial resolution in three upsampling blocks, with each block received skip connections from the corresponding encoder level.

Because paired high-resolution data were absent across different anatomical planes, we regulated the self-supervised learning process with two key objectives (**Figure 1b**). The first objective was the use of the contrastive PatchNCE loss to maximize the deep feature similarity between the input and enhanced volumes, maintaining the fidelity of the features at multiple scales. The PatchNCE loss enabled patch-level contrastive learning to be performed for translating unpaired images by increasing the amount of mutual information between the corresponding patches in the source and translated images (Park et al., 2020). When we translated an input patch *x* from the low-resolution coronal and axial view to a high-resolution version, we calculated the feature losses induced by ℓ layers $\{\ell_1 \ldots \ell_L\}$ using the encoder part of the network. At each layer ℓ, the network sampled *S* positive patches and *N* negative patches, yielding

$$L_{patchNCE}(G, x) = \sum_{\ell} \sum_{s=1}^{S} \ell(v_s^{\ell}, v_s^{'\ell}, \{v_n^{\ell}\}_{n=1}^{N}) \qquad (1)$$

where v_s^{ℓ} and v'^{ℓ}_s are the features of the low-resolution and enhanced high-resolution patches, respectively, and $\{v_s^{\ell}\}_{n=1}^{N}$ denotes the negative sample features. The InfoNCE loss for each patch pair is

$$\ell(v^{\ell}, v'^{\ell}, \{v_n^{\ell}\}) = -\log\left[\frac{exp(v \cdot v'/\tau)}{exp(v \cdot v'/\tau) + \sum_{n=1}^{N} exp(v \cdot v_n/\tau)}\right] \qquad (2)$$

with a temperature parameter τ. This objective encouraged the network to learn domain-invariant features across multiple scales while preserving the structural correspondence between the source and translated images without requiring paired training images.

A multi-view adversarial loss was utilized to ensure the presence of consistent high-resolution characteristics across all enhanced image planes through discriminator-based adversarial loss. While the sagittal view provided high-resolution information, the discriminators minimized the significant differences between this view and the enhanced coronal and axial projections of the generated volume by optimizing the following adversarial losses:

$$L_{adv}^{\mathrm{cor}}(G, D) = \mathrm{E}[D_{\mathrm{cor}}(y_{\mathrm{sag}}) - 1] + \mathrm{E}[D_{\mathrm{cor}}(G(x)_{\mathrm{cor}})] \tag{3}$$

$$L_{adv}^{\mathrm{ax}}(G, D) = \mathrm{E}[D_{\mathrm{ax}}(y_{\mathrm{sag}}) - 1] + \mathrm{E}[D_{\mathrm{ax}}(G(x)_{\mathrm{ax}})], \tag{4}$$

where D denotes the discriminator network minimizing the difference between the high-resolution sagittal view (Y_{sag}) and the generated coronal and axial views, with the superscripts indicating the slice directions on the coronal and axial views, respectively. Adversarial training alternated between the coronal and axial discriminators, providing improved stability and maintaining consistent texture quality across all views. Additionally, unlike some previously developed methods that rely on cyclic losses derived from an inverse generator, our method used a simpler L_1 loss for comparing the original MR image with maximum-intensity projections (sampled every n voxels along the sagittal axis of the enhanced volume, matching the anisotropic ratio of the input image volume), as follows:

$$L_1 = \frac{1}{N}\sum_{ij} |x(i, j, 8k) - max_k y(i, j, k)|. \tag{5}$$

This step maintains the fidelity of the input structure while providing the model with the flexibility to enhance its output quality through adversarial training. The final target of the optimization process L for the enhancement model can be summarized as

$$L = L_{adv}^{\mathrm{cor}}(G, D) + L_{adv}^{\mathrm{ax}}(G, D) + \alpha L_{patchNCE}(G, X) + \beta L_1 \tag{6}$$

where $\alpha = 0.1$ and $\beta = 10$ are weighting parameters. During the model training procedure, we implemented a cosine learning rate schedule using an adaptive moment estimation (Adam) optimizer with an initial learning rate of 5×10^{-5} that was maintained across 1000 epochs with a batch size of 1 to accommodate memory constraints. The GAN architecture featured an encoder-decoder generator with skip connections and a 2D patch-based discriminator (Zhu et al., 2017) operating on the axial and coronal planes. Owing to the memory limitations of the GPU for handling 3D images, the MR images were processed as overlapping patches consisting of 128^3 voxels with a stride of 64 voxels to ensure consistent reconstruction effects across the volume boundaries.

2.4 Identification of 3D Image Abnormalities

An anatomy conditioned denoising diffusion model identified image abnormalities by transforming each diseased MR image into a healthy-appearing MR image while preserving its normal anatomy (**Figure 1c**). To localize abnormal regions and guide the denoising process, we initialized the diffusion model with a coarse abnormal map *m*, which was derived from our prior work (Lian et al., 2023), to produce a mask that highlighted the candidate regions containing abnormalities. Formally, let $m(i, j, k) \in \{0, 1\}$ indicate whether voxel (i, j, k) is designated as "abnormal" (*m*=1) or "healthy" (*m*=0). (Ho et al., 2020)Let x_o be the original MR image. Inside the abnormal map, we applied progressive Gaussian noise to obscure the potential abnormalities, thus creating a partial observation. The forward process at step *t* can therefore be formulated as [38]

$$x_t = (1 - m) \odot x_o + m \odot \left(\sqrt{\overline{\alpha_t}} \cdot x_o + \sqrt{1 - \overline{\alpha_t}} \cdot \epsilon_t\right), \tag{7}$$

where $\odot$ denotes the elementwise multiplication operation, x_t is the image at diffusion step t, $\overline{\alpha_t}$ is the noise scheduling parameter, and $\epsilon_t \sim \mathcal{N}_t(0, \mathrm{I})$ is a sample acquired from a normal distribution. Conversely, the backward process involved estimating the posterior distribution of the less-noisy image given a noisy observation:

$$p_\theta(x_{t-1}|x_t, m) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t, m), \Sigma_\theta(x_t, t, m)\right), \forall t \in 1, \ldots, T, \tag{8}$$

where μ_θ and Σ_θ are the mean and variance, respectively, of the noisy image, as predicted by a U-Net-based denoising network with a parameter θ, and the final healthy image can be reconstructed through iterative denoising as follows:

$$p_\theta(x_{0:T}, m) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t, m). \tag{9}$$

To further preserve the spatial structure during denoising, we integrated spatially adaptive normalization (SPADE) blocks (Park et al., 2019) into the denoising model. In each layer of the U-Net denoising encoder, SPADE transformed the normalization parameters learned from the anatomical segmentation map, s:

$$h_\ell \leftarrow \mathrm{Conv}(\mathrm{SPADE}(h_{\ell-1}, s)) \tag{10}$$

where h_ℓ is the feature map at layer ℓ. Because the spatial modulation process depends explicitly on anatomical boundaries, the denoising process was better constrained to maintain anatomical consistency in the healthy regions while focusing the reconstruction capacity of the model on the masked abnormal areas. Additionally, we incorporated an adjacent-information encoder (AIE), which provided cross-slice continuity when synthesizing the 3D volumes. The AIE shared the same model architecture as that of the encoder part of the denoising model. Features acquired from nearby slices (x_{n+1}) were encoded and injected into the denoising network via channel-wise multiplication. This mechanism preserved the structural coherence between the slices and prevented unintended alterations in the healthy tissue outside the diseased region.

During the model inference process**,** the diffusion model converted an MRI scan with disease to a “healthy” version of the scan inside the abnormal mask (m=1) while retaining the image intensities outside of this region. The final 3D abnormality map was obtained by differentiating the diseased MR images and the generated healthy images, both of which were enhanced to isotropic resolutions by the isotropic enhancement model.

2.5 Model implementation

The isotropic enhancement GAN adopted a 3D U-Net–style encoder–decoder (32 base channels, skip connections, batch normalization, rectified linear unit (ReLU); linear output) and a 16 × 16 patch discriminator; both were Xavier-initialized and trained for 2 000 epochs plus a 100-epoch cosine-decay tail using Adam (lr = 5×10^{-5}, $\beta_1 = 0.5$) with random ± 45° rotations, gradient clipping and a cross-entropy adversarial loss combined with a max-projection L_1 term (weight = 1). Model checkpoints were saved every 100 epochs on eight NVIDIA A6000 GPUs. For abnormalities localization, we utilized the Palette diffusion inpainting network: a Kaiming-initialized 256 × 256 U-Net (two residual blocks per scale, attention at a feature resolution of 16, and a dropout rate of 0.2) trained with EMA (2000 and 1000 steps for training and testing, respectively) under an Adam optimizer (lr = 5×10^{-5}, batch = 4). During inference, 200 forward-diffusion steps masked the suspected osteoarthritic voxels, and reverse denoising was employed to reconstruct their healthy counterfactuals. The comprehensive implementation details, network architecture specifications, and hyperparameter configurations are provided in Supplementary Material B. For transparency and reproducibility, we have released all training and evaluation codes, sample data, and model checkpoints online (github.com/MicroCoreNTU/IsotropicOAI).

2.6 Performance metrics and quality evaluation

We evaluated the enhanced isotropic MRI volumes and image anomalies using the hold-out validation dataset with separate subjects. Given the self-supervised nature of our study and the absence of high-resolution IW-TSE MR images from the OAI imaging cohorts, the quality of each generated MR volume was measured on the basis of several image quality quantifiers used for assessing the realism and feature similarities between the sagittal, coronal, and axial views of the generated MR volume and the real MR images. These quality measures included the Fréchet inception distance (FID), KID, and perceptual similarity assessment metrics, such as the LPIPS. To evaluate the degree to which anatomical structures were preserved between the synthesized and original MR images, we employed a previously validated automatic bone segmentation model to compare the anatomical segmentation results produced for the femur and tibia bone areas on the enhanced coronal and axial views. The structural consistency level was quantified using the F1 score, which provided a standardized spatial overlap measure ranging from 0 (no overlap) to 1 (complete overlap). An expert orthopedist with advanced experience involving musculoskeletal MR images evaluated the image quality levels and confirmed the detected anomalies by examining the generated MR images.

3. Results

3.1 Isotropically enhanced visualization

Comparisons between the enhanced isotropic MR images and their anisotropic counterparts revealed substantial cross-sectional visualization improvements within the enhanced coronal and axial planes (**Figure 2a**). The proposed method significantly improved the fidelity of the images while preserving the anatomical structures of the original versions, thereby mitigating the discontinuity artifacts that are typically introduced by interslice interpolation in MR images with large slice thicknesses. The expert evaluation further confirmed the notable abnormal feature enhancements across the key anatomical regions, including horizontal meniscal tears, collateral ligament morphology, patellar effusion, and bone marrow lesions (**Figure 2b**).

3.2 Expert preference evaluation

To evaluate the perceptual quality of the synthesized MR images, we conducted a blinded preference study involving an expert with extensive musculoskeletal MRI knowledge. The assessment involved pairwise comparisons between the reconstructions yielded by competing models and the source MRI data, with a focus on the overall image quality and anatomical preservation levels. Our proposed model was preferred in 65 % of the axial evaluations and 67 % of the coronal evaluations (**Figure 3a**), outperforming the existing methods. Although adding cyclic consistency to the model yielded the second-highest preference score, the corresponding images exhibited evident structural distortion (**Figure 3b**). By omitting the backward generation network, which is commonly employed in other self-supervised superresolution frameworks, our model delivered superior image quality while maintaining the original anatomical structures with minimal image artifacts.

3.3 Quantitative validation and cross-site generalization

The quantitative analysis results further reinforced these findings. As depicted in **Figure 4**, the proposed method consistently yielded the lowest FID, KID, and LPIPS scores across both the axial and coronal planes, indicating superior perceptual realism and feature-level fidelity (**4a**). Relative to those of the anisotropic source volumes, the FIDs decreased from 407.4 to 254.4 in the coronal view and from 429.9 to 266.9 in the axial view. The KID and LPIPS showed similar reductions, demonstrating both improved distributional alignment and perceptual similarity relative to the high-resolution references when compared to state-of-the-art self-supervised models (**4b**). Additionally, to assess whether this enhanced isotropic resolution translated to downstream tasks, we quantified the effect of resolution enhancement on the automated bone segmentation process . **Figure 4c** shows markedly more accurate bone delineation effects in the coronal plane, particularly along the complex femorotibial interface. The enhanced images contained smoother, anatomically accurate articular surfaces with substantially less interslice discontinuity than those derived from the original inputs. When benchmarked against manually annotated ground truths acquired from 30 subjects, the segmentations derived from the enhanced volumes demonstrated significantly higher geometric accuracy, as evidenced by a reduced average surface distance to the referenced annotations. Additionally, to evaluate the generalizability of the resolution enhancement module, we conducted a series of validations. As illustrated in **Figure 5**, the model consistently improved image quality across five different sites, maintained stability under synthetic noise and contrast perturbations, and successfully generalized to diverse MRI sequences, including T2-SPACE spine and DESS knee imaging.

3.4 Enhanced 3D abnormality detection and morphological precision

Next, we investigated whether the improved image resolution could facilitate a more accurate 3D abnormality detection process when combined with our diffusion module. In the sagittal view, the diffusion model selectively removed abnormal tissues while preserving surrounding normal structures (**Figure 6a**). Following the resolution enhancement procedure, the detected abnormalities were further sharpened in the 3D space: multiple bone marrow lesions that appeared confluent at the native resolution were resolved into discrete instances, and the infrapatellar abnormalities along the bone surface were delineated with higher precision (**Figure 6b**). A frequency-domain analysis also reflected these volumetric improvements; the spectrum derived from the native-resolution abnormalities displayed prominent directional artifacts along the sagittal axis, whereas the spectrum obtained from the enhanced volumes was radially symmetric (**Figure 6c**). These findings also translated to a more reliable downstream procedure: the enhanced bone-segmentation masks rectified the previous bone-versus-soft-tissue misclassifications (arrow, **Figure 6d**) and enabled more precise 3D abnormalities attribution. A quantitative morphology analysis further substantiated these gains; the compactness scores obtained for bone marrow lesions and effusion masks increased significantly after applying the enhancement, reflecting smoother, less fragmented boundaries (**Figure 6e**).

3.5 Ablation study of AIE and SPADE module

To clarify the individual contributions of the AIE and the SPADE blocks within the diffusion-based abnormality detection network, we performed a systematic ablation study on coronal reconstructions (**Figure 7a**). The full model—AIE + SPADE—successfully localized 3D abnormalities while preserving anatomical structures with the source MR images and maintaining continuity across the sagittal slices. In contrast, removing the AIE disrupted the interslice coherence, producing visible discontinuities within effusion regions and distorting the bone shapes. Eliminating the SPADE blocks (while retaining the AIE) preserved continuity but introduced significant shape distortions, indicating the role of SPADE in enforcing slice-wise structural fidelity. These observations justify the integrated use of both modules in the final diffusion model.

4. Discussion

In this study, we propose a novel approach that integrates a probabilistic diffusion model with a self-supervised resolution enhancement framework to reconstruct 3D structures and detect abnormalities from anisotropic MR images. Anisotropic MRI resolutions often impede the initial diagnosis and image analysis tasks, including longitudinal assessments of image abnormalities. Although expert radiologists can effectively interpret anisotropic data, computerized analytical methods such as abnormality detection and segmentation typically require high-resolution isotropic images. Our model enhanced the images of knee structures and detected abnormalities in anisotropic MR images, enabling the more precise volumetric quantification of key imaging biomarkers. Importantly, our method did not require high-resolution coronal or axial knee MR images to provide these outputs, making it readily applicable to single-view MRI scans without ground-truth isotropic data, which is a significant advantage given the difficulty of acquiring such references. The reconstruction quality improvements achieved by our model were further confirmed through image quantitative metrics such as the KID and LPIPS, and the accuracy of its ability to detect abnormalities was confirmed by an expert review. Together, these results demonstrate that image analysis tasks beyond segmentation can benefit from self-supervised resolution enhancement methods because they reveal previously underresolved information in anisotropic MR images. This ability should benefit longitudinal studies of the existing cohorts, where the acquisition parameters cannot be modified.

Our model demonstrated a better ability to perform self-supervised resolution enhancement than the previously reported methods did without relying on a secondary generator or cyclic consistency constraints. The present framework exhibited improved efficiency in terms of enhancing MR images, requiring a single generator model while avoiding the propagation of errors through backward generators. In our experiments, introducing a cyclic loss led to noticeable artifacts and structural inconsistencies across different anatomical planes. By excluding these components and using patch-based contrastive learning rather than pixelwise reconstruction, our approach reduced such artifacts while improving its computational efficiency. In addition, the model handled high anisotropic ratios (approximately 8:1) well, surpassing the capabilities of the existing methods, which are typically limited to 2–4× factors. This improvement can extend the clinical applicability of the proposed method to a wider range of MRI acquisitions that are commonly used in routine musculoskeletal imaging protocols.

Despite these promising results, the absence of truly isotropic ground-truth data means that we could not definitively confirm the voxel-level fidelity of our outputs. Our reliance on surrogate metrics (the FID, the KID, and LPIPS) and segmentation-based assessments, while informative, cannot eliminate potential over-smoothing or hallucination when addressing subtle structures. A future extension of this study would involve small-scale phantom experiments or a limited cohort scanned with near-isotropic protocols, providing a more direct quantitative benchmark. Nevertheless, expert reviews have confirmed the markedly improved visibility of the bone marrow lesions and effusion-synovitis in previously low-resolution planes, indicating the ability of our framework to enhance the 3D abnormality detection process. By generating isotropic 3D reconstructions that distinctly separate abnormalities from healthy tissues, our method enables more accurate volumetric assessments and clearer delineations of disease-related changes. In future longitudinal investigations, these enhancements could also facilitate multiplane comparisons across different timepoints, mitigating the alignment issues that arise from highly anisotropic acquisitions.

5. Conclusion

In conclusion, the proposed self-supervised pipeline offers immediate practical value by converting fast but anisotropic IW-TSE sequences into diagnostically robust, fully isotropic volumes. This capability can supplant lengthier 3D fast-spin-echo acquisitions, trimming minutes from routine knee MRI scanning processes while reducing motion artifact risks and improving the overall clinical throughput. The orthopedist involved in our blinded study consistently preferred the enhanced images, citing their sharper depictions of bone marrow lesions, meniscal tears and effusion–synovitis; the use of a uniform voxel geometry also prevented subtle osteoarthritic changes from vanishing when viewing by scrolling, which is a benefit that is likely to provide increased inter-reader agreement for scoring systems such as WORMS and MOAKS. Beyond day-to-day reporting, isotropic synthesis unlocks the vast stores of highly anisotropic scans contained in large-scale cohorts—more than 10,000 in the OAI alone—enabling retrospective 3D morphometry and longitudinal biomarker discovery studies without the need for paired high-resolution ground truths. Importantly, the enhanced image quality is translated into lower anatomical segmentation and abnormalities attribution errors, indicating a broader downstream gains for tasks ranging from cartilage thickness mapping to multimodal registration, indicating broader clinical applications.

CRediT authorship contribution statement

G.H. Chang: Conceptualization, Writing – Original Draft, Supervision, Project Administration, Writing – Review & Editing, Final Approval. **J.Y. Hsu**: Methodology, , Investigation, Validation, Visualization, Writing – Review & Editing, Final Approval. **P.H. Lian**: Methodology, Software, Investigation, Writing – Original Draft, Writing – Review & Editing, Final Approval. **Y.H. Chen:** Methodology, Software, Investigation, Visualization, Final Approval.**T.Y. Chuang**: Methodology, Software, Investigation, Visualization, Final Approval.

Declaration of competing interests

The authors report no conflicts of interest.

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Declaration of generative AI in scientific writing

During the preparation of this work, the authors used Claude Sonnet to assist with checking the grammar of the manuscript and refining its language. After using this tool/service, the authors reviewed and edited the content as needed, and they take full responsibility for the content of the published article.

Data statement

The MRI datasets used in this research were obtained from the publicly available Osteoarthritis Initiative (OAI), accessible to qualified researchers through the National Institute of Mental Health Data Archive (https://nda.nih.gov/oai). All model training and evaluation scripts, sample datasets, and model checkpoints developed in this study are publicly accessible at github.com/MicroCoreNTU/IsotropicOAI. This repository includes training and inference scripts with environment setup instructions.

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Figure 1

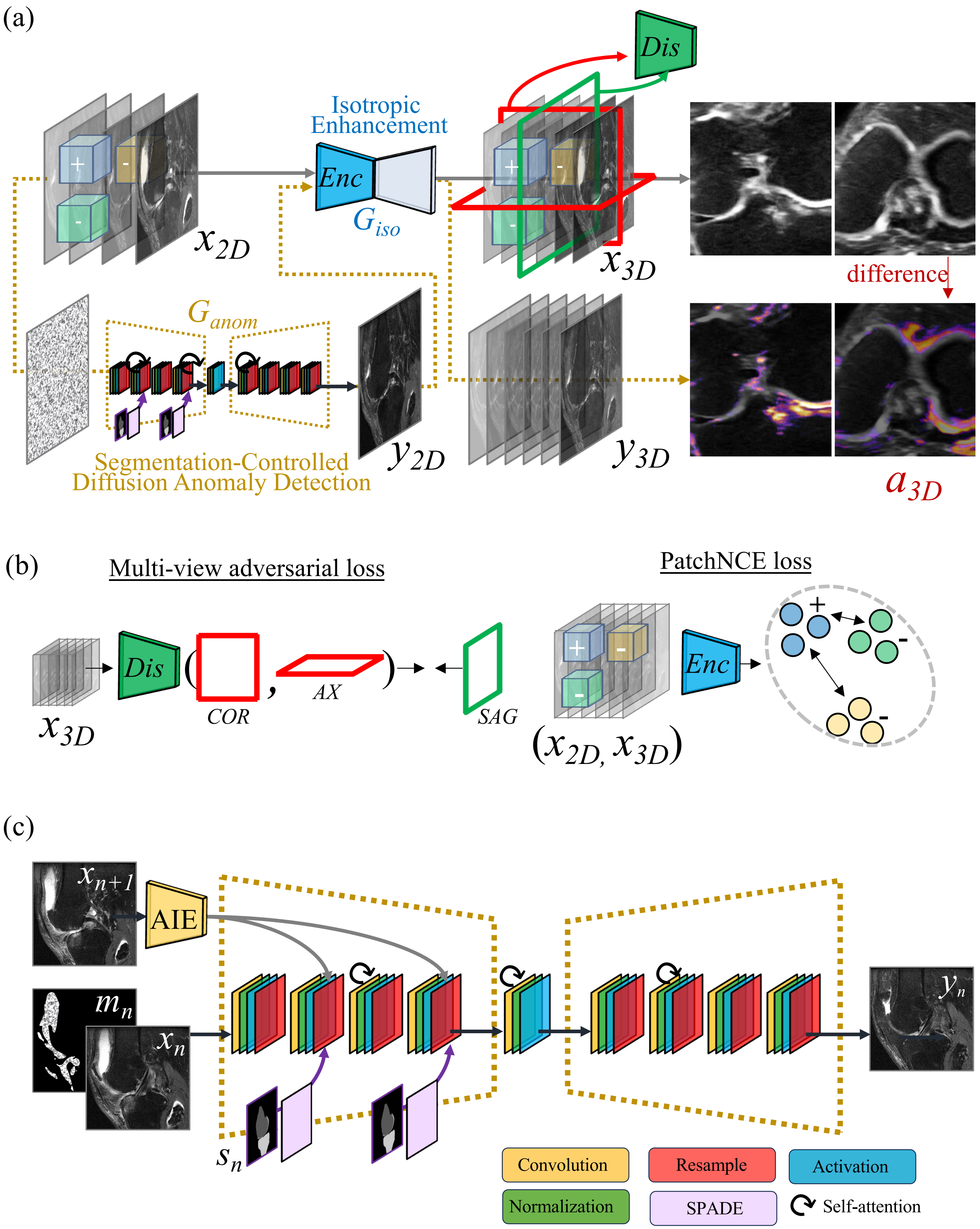
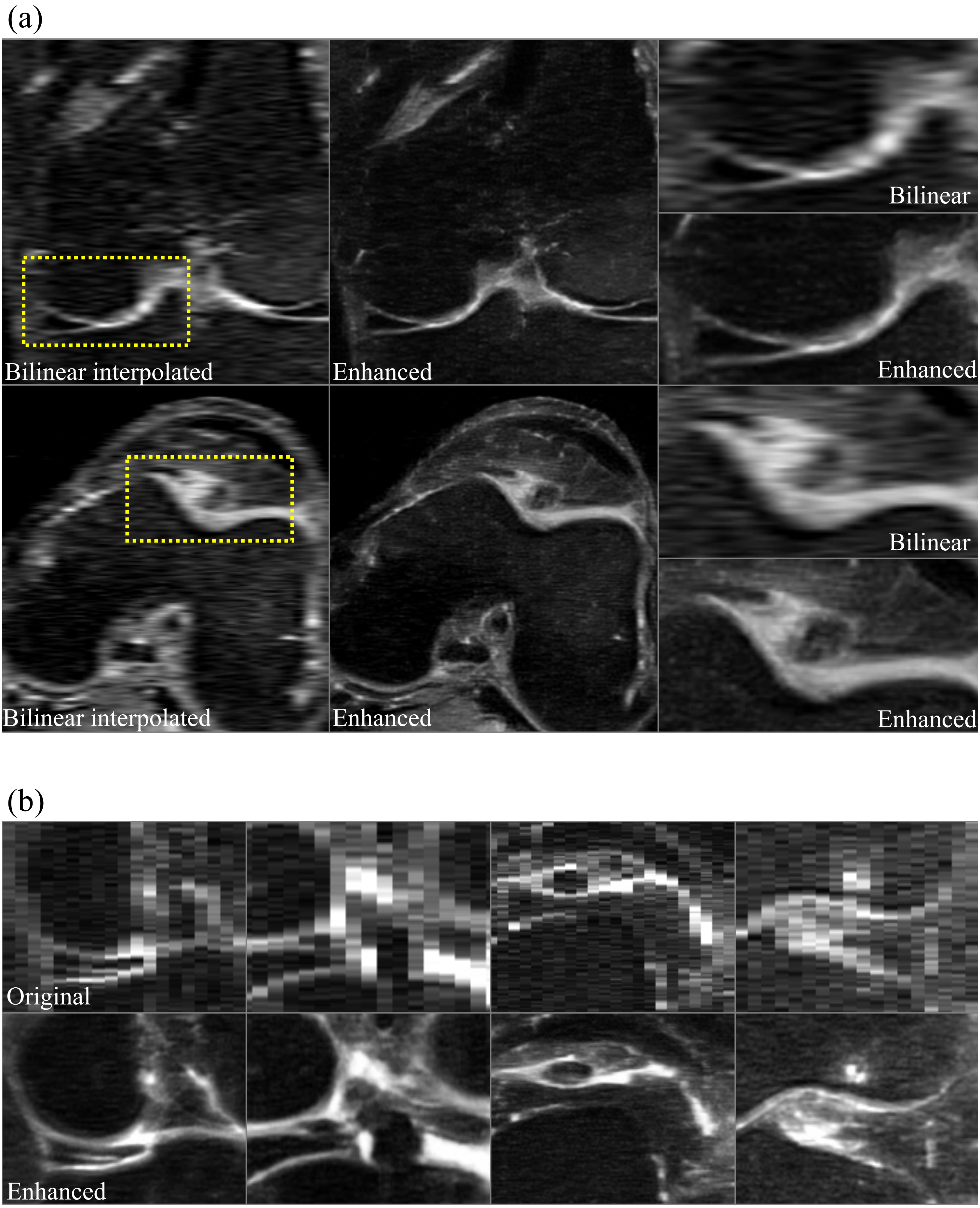

(a)
Isotropic Enhancement
Enc
Dis
G_{iso}
x_{2D}
x_{3D}
difference
G_{anom}
y_{2D}
y_{3D}
a_{3D}
Segmentation-Controlled Diffusion Anomaly Detection
(b)
Multi-view adversarial loss
PatchNCE loss
x_{3D}
Dis
COR
AX
SAG
Enc
(x_{2D}, x_{3D})
(c)
x_{n+1}
AIE
m_n
x_n
s_n
y_n
Convolution
Resample
Activation
Normalization
SPADE
Self-attention

Figure 2

Click here to access/download;Figure(s);fig_2.pdf



(a)

	Expert Preference Score	
Model	Axial View	Coronal View
Bilinear	0.00	0.00
[20]	0.00	0.01
[25]	0.22	0.29
Full Model (ours)	**0.65**	**0.67**
w/o CL (ours)	0.06	0.10

(b)

Original | Bilinear | [20] | [25] | **Full Model** | w/o CL

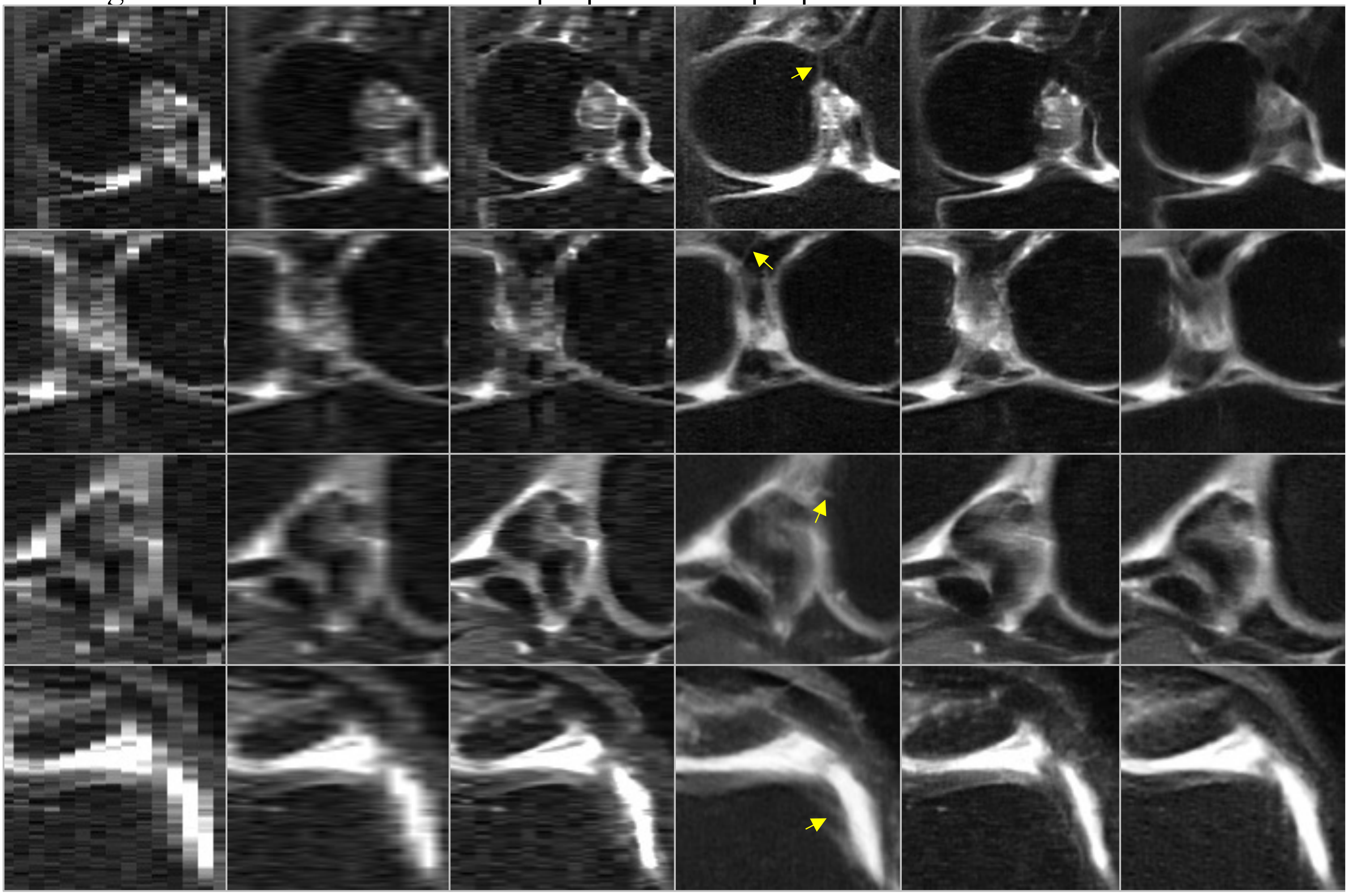
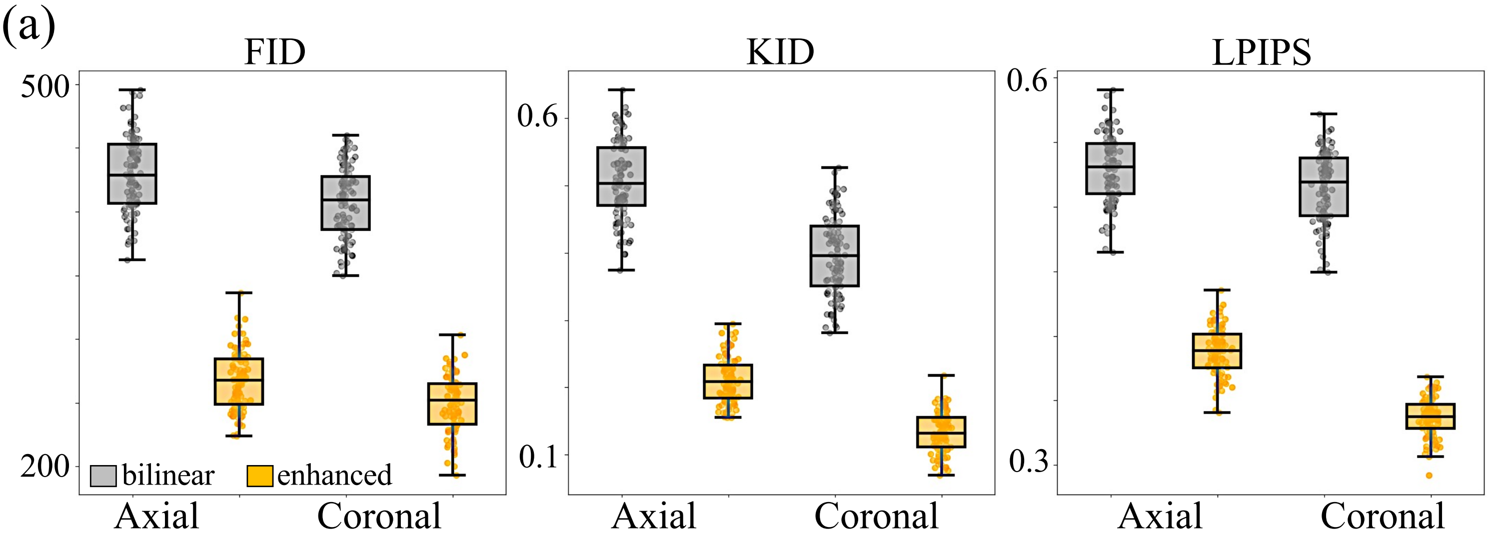


Figure 4 Click here to access/download;Figure(s);fig_4.pdf

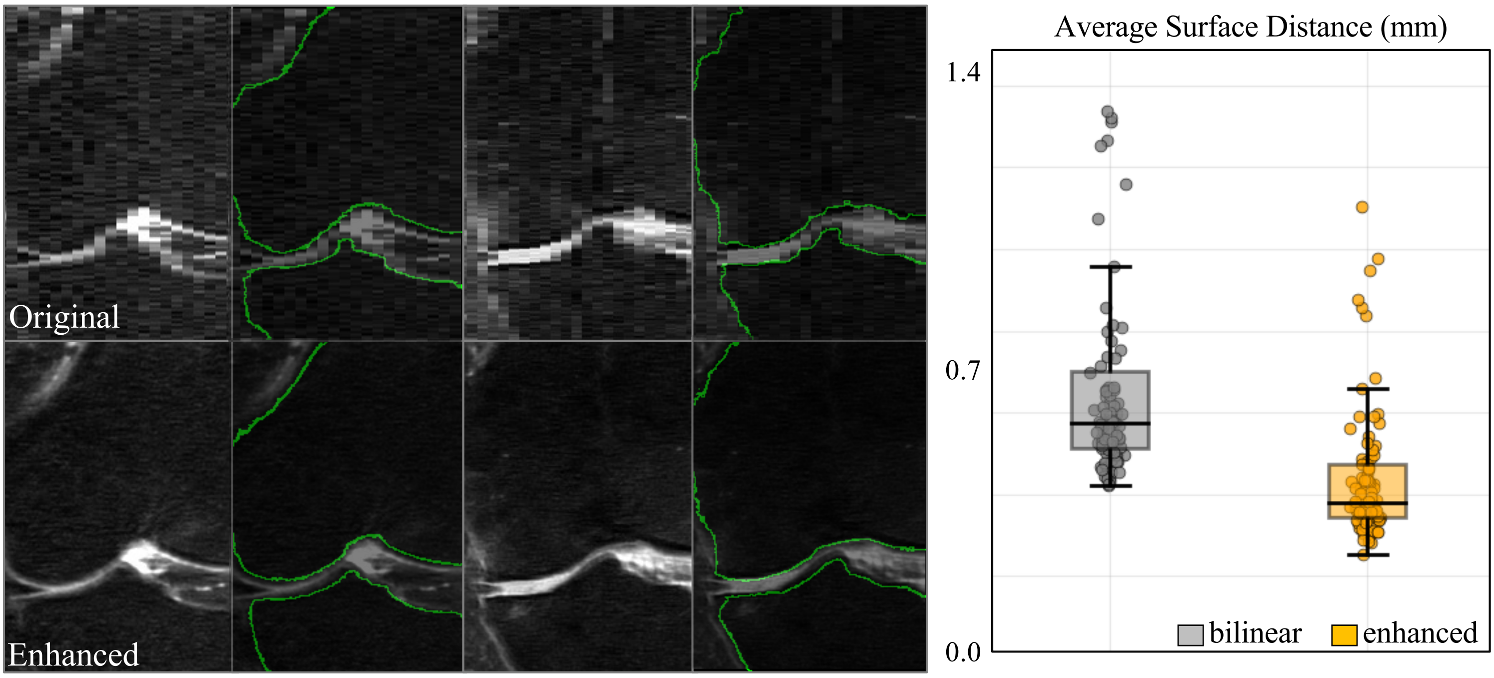
(a)



(b)

	Coronal			Axial		
Methods \ Metrics	FID↓	KID↓	LPIPS↓	FID↓	KID↓	LPIPS↓
Original	407.4±27.9	0.396±0.06	0.516±0.03	429.9±29.5	0.508±0.06	0.530±0.03
Bilinear	349.6±33.1	0.306±0.07	0.490±0.02	342.1±32.3	0.341±0.05	0.504±0.03
[20]	310.9±39.4	0.250±0.07	0.437±0.03	293.4±26.9	0.258±0.04	0.456±0.03
[25]	278.1±18.8	0.165±0.03	0.367±0.01	289.6±25.0	0.232±0.04	0.406±0.02
Proposed	**254.4±25.4**	**0.133±0.03**	**0.337±0.01**	**266.9±22.8**	**0.211±0.03**	**0.387±0.03**

(c)



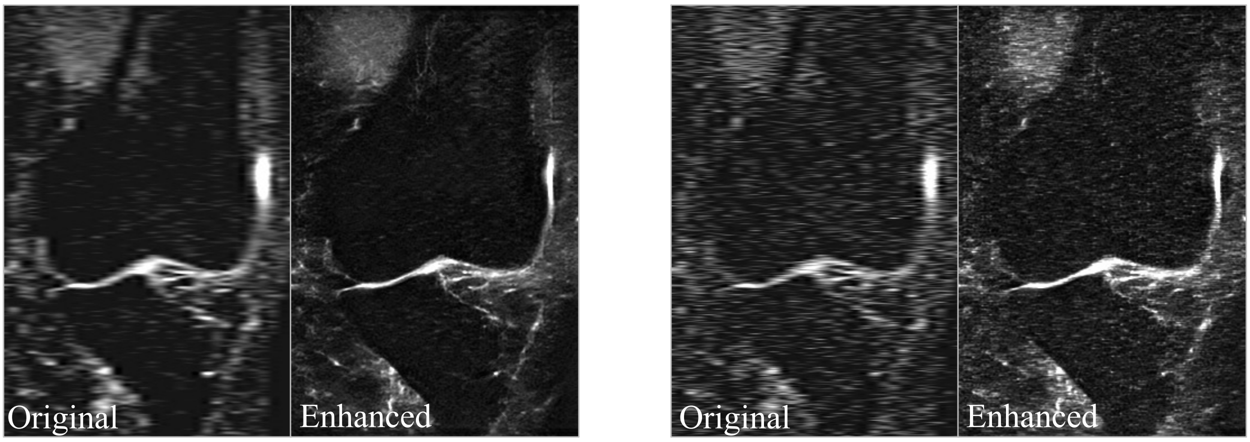
(a)

	Coronal			Axial		
Sites \ Metrics	FID↓	KID↓	LPIPS↓	FID↓	KID↓	LPIPS↓
A	251.8 ± 10.0	0.143 ± 0.02	0.342 ± 0.01	274.1 ± 12.7	0.230 ± 0.03	0.387 ± 0.01
B	257.5 ± 19.3	0.139 ± 0.02	0.340 ± 0.02	278.7 ± 28.0	0.219 ± 0.04	0.393 ± 0.02
C	251.1 ± 18.4	0.134 ± 0.02	0.336 ± 0.01	261.9 ± 18.2	0.206 ± 0.03	0.386 ± 0.02
D	239.1 ± 21.5	0.120 ± 0.03	0.332 ± 0.01	265.0 ± 24.7	0.208 ± 0.04	0.384 ± 0.02
E	260.2 ± 30.5	0.152 ± 0.04	0.345 ± 0.02	264.9 ± 16.6	0.215 ± 0.02	0.395 ± 0.02

(b)

Brightness & contrast perturbation

Gaussian & multiplicative noise



	Coronal			Axial		
Methods \ Metrics	FID↓	KID↓	LPIPS↓	FID↓	KID↓	LPIPS↓
Bilinear	368.0 ± 38.7	0.375 ± 0.08	0.494 ± 0.04	339.2 ± 28.6	0.368 ± 0.05	0.512 ± 0.04
Proposed	252.1 ± 28.1	0.191 ± 0.05	0.375 ± 0.03	270.4 ± 23.4	0.256 ± 0.04	0.416 ± 0.03

(c)

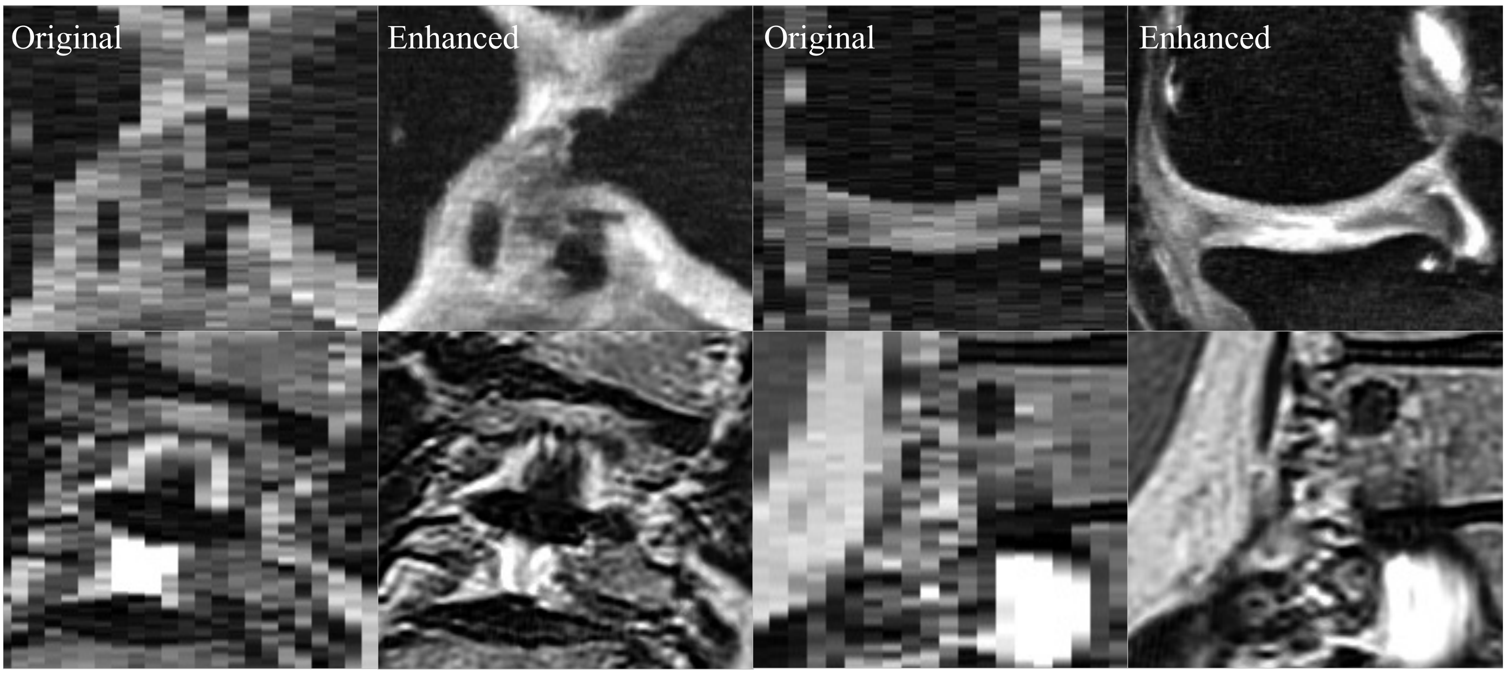


Figure 6

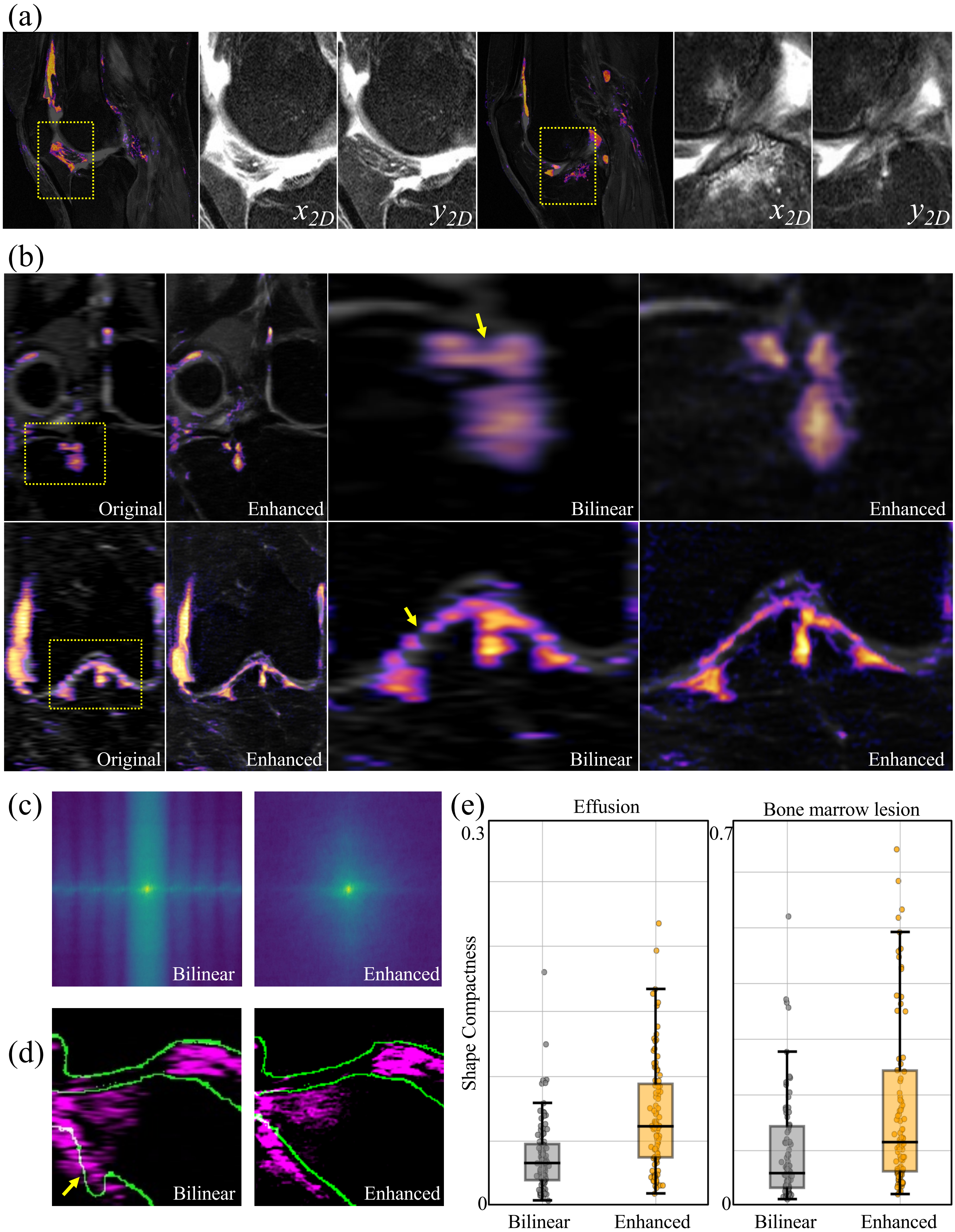
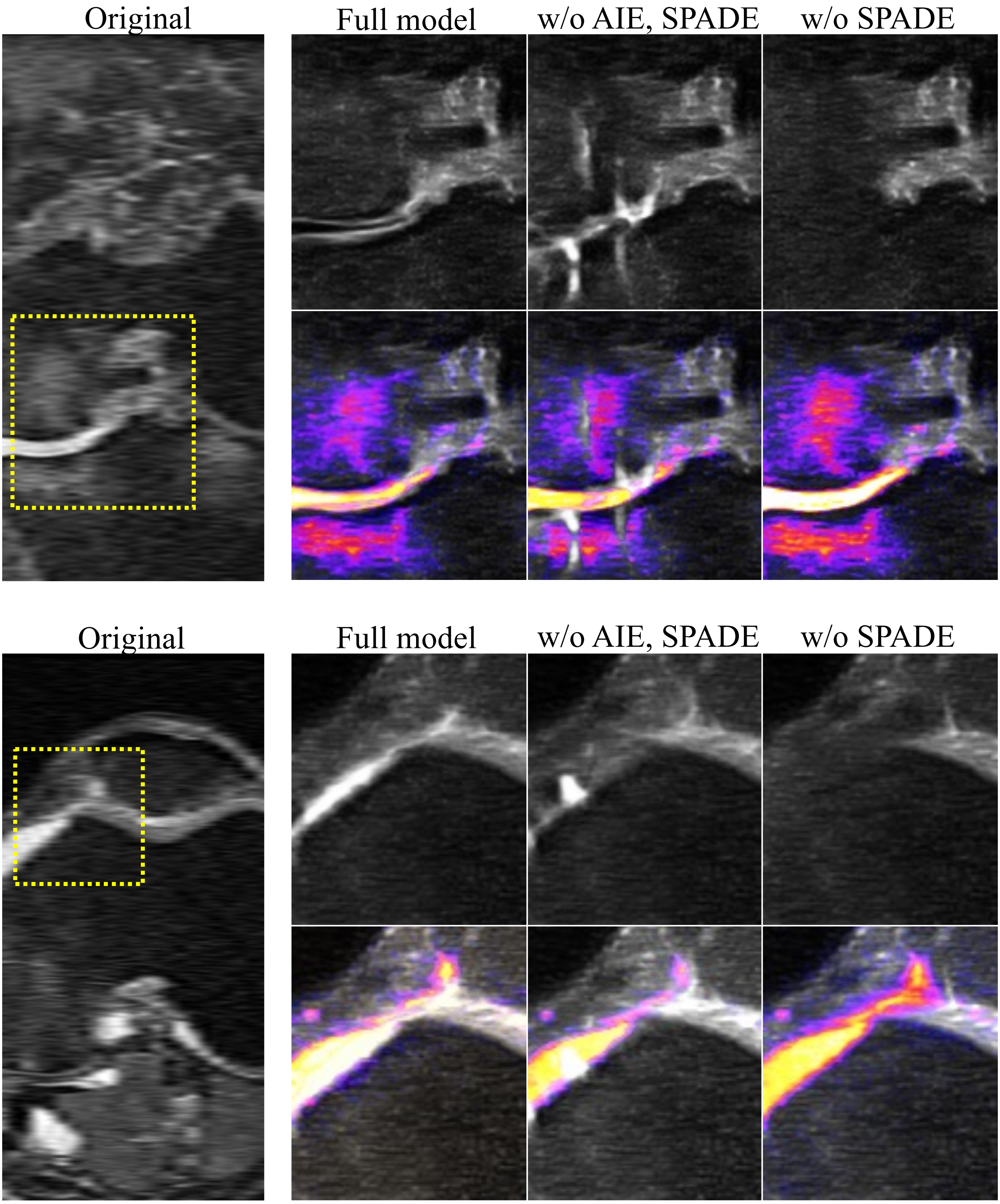

(a)
x_{2D}
y_{2D}
x_{2D}
y_{2D}
(b)
Original
Enhanced
Bilinear
Enhanced
Original
Enhanced
Bilinear
Enhanced
(c)
Bilinear
Enhanced
(d)
Bilinear
Enhanced
(e)
Effusion
Bone marrow lesion
0.3
0.7
0
0
Shape Compactness
Bilinear
Enhanced
Bilinear
Enhanced

Figure 7


Original
Full model
w/o AIE, SPADE
w/o SPADE
Original
Full model
w/o AIE, SPADE
w/o SPADE